Predicting a deflection from an impact

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Estimate a velocity change and explain the value of lead time.

SUCCESS CRITERIA - I can...

1. State how you simplify the collision.
2. Connect delivered momentum to target velocity change.
3. Explain why real deflection includes ejecta and orbital effects.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A **0.3 kg** projectile at **50 m/s** sticks to a stationary **2.7 kg** cart. **Find common speed** and the fraction of initial kinetic energy remaining.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\Delta v \approx p_{projectile}/M_{target}$; shift ≈ Δv × lead time. Simplified calculation: target mass is much larger; straight-line shift comparison. Ejecta can also transfer momentum in a real impact.

Worked reasoning: In a fictional collision, a **2 kg** projectile at **30 m/s** sticks to a stationary **10 kg** target. Final speed is **60/12=5 m/s**. The system includes both objects.

Your turn: Draw the before-and-after system boundary and momentum arrows.

Before **After**

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

After a brief sideways push, a probe has a constant added sideways velocity. Assume no further sideways force. **Compare its sideways distance after** a short drift and a long drift. **Use the diagram** to connect impulse, velocity change, and later position.

Impulse J → Change Δv → Offset Δx

Marzano check - circle: ① ② ③

04 Apply the physics

A hypothetical **600 kg** impactor moving at **6000 m/s** sticks to a **5 × 10⁹ kg** target, initially at rest in the target frame. Ignore ejecta. **Estimate the target's velocity change** and straight-line offset after **10⁶ s**. **Label the system before** and after impact. These are values for this practice calculation, not mission measurements.

Before **After**

Label objects and signed velocities.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Why should the no-ejecta sticking calculation not be reported as the exact measured result of a real mission?

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

An asteroid's crossing window is days **16**-**20**; Earth's is **18**-**22**. **Find the overlap** and **explain why this** is not proof of a collision.
Organize the timing information before drawing a conclusion.

Object	Arrival time / window	Comparison
A	____	____
B	____	____

Marzano check - circle: ① ② ③

07 Exit check

A hypothetical target needs a **10 km** shift over **2×10^{6} s**. **Find required** Δv. If target mass **10^{9} kg**, estimate impulse. Organize the known quantities on the before-and-after diagram.

Before | **After**

Label objects and signed velocities.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A **1000 kg** hypothetical target needs a transverse velocity change **0.02 m/s**. **Estimate required impulse**. If delivered uniformly for **0.1 s**, find average force and displacement from the velocity change after **10^{5} s** in a simplified calculation with no further force. Organize the known quantities on the before-and-after diagram.

Before | **After**

Label objects and signed velocities.

Marzano check - circle: ① ② ③